

...and watch your game become a movie!



Version 0.91b

User Guide and Manual

For the version 0.92b release





General Information

“Cinematic Perspective” is a Unity asset that has the objective of providing the developer with a simple, intuitive and fast way to implement basic camera rig movement, focal and perspective automation as it has been extensively used by filmmakers in the industry to provide a more interesting picture composition or provide meaning to a shot.

The natural use case scenario for this asset varies. It can be implemented for cutscenes, it can be used in gameplay (as in classic games like LucasArts “Grim Fandango”), also it can be used as an interesting way to show architectural design in unity, in hybrid solutions, and various other ways. Creative imagination is the limit.

Please be advised that at the moment of the confection of this manual is still in beta phase. Main testing has been done in Windows 10 Platform under Unity 5.5.

All inquiries should be directed to: support@cinematicperspective.com

The help forum is: <https://forum.unity3d.com/threads/cinematic-perspective-beta-0-9.454274/>

The asset website is: <http://www.cinematicperspective.com/>

Asset Store Link: <https://www.assetstore.unity3d.com/en/#!/content/82054>

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Introduction

The “Cinematic Perspective” asset is a class that allows the operation of a set of camera rigs. Camera rigs are virtual devices used to position, orientate and move a cameras based on four modes.

The Steadicam mode (or Follow shot or tracking shot) is a specific camera angle in which the subject being filmed is seemingly pursued by the camera, for example by a Steadicam. This particular mode will follow the target or subject (performer) in motion.

The LookAt mode is a mode where the rig remains in a fixed position, but the camera faces the target and the Zoom (FOV) can be automated or manually adjusted. **The Fixed mode** is a mode where the rig remains in a fixed position and the camera faces a defined direction. In this mode the Zoom (FOV) can be manually adjusted. **The Dolly mode** basically moves the camera independent from the target or subject (performer). The camera can face the target, or another target, or simply focus in a particular direction. The movement is in a vector from point a to b and the movement speed can be edited in an animation curve. Since version 0.92b transitions can be used.

All of this modes are used in the demo scene found in the example folder of the asset.

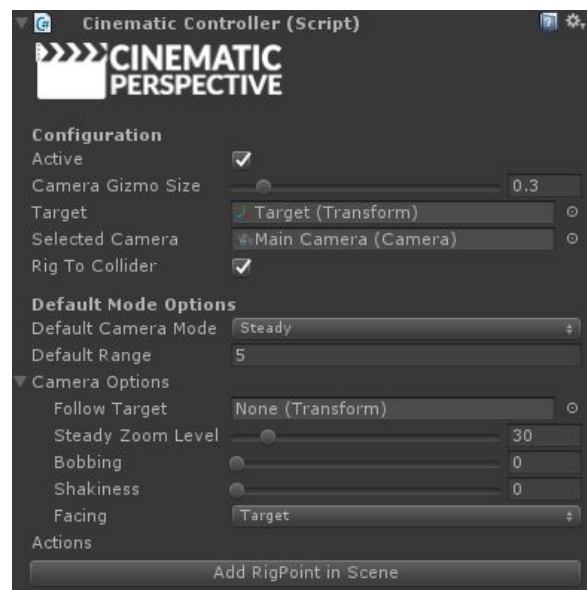
Getting Started

The “Cinematic Perspective” asset is provided with a main prefab found directly under the main folder.

Important: this asset has been coded to work in the “Assets/CinematicPerspectives/” folder of your project. In case this has to be moved, please adjust the code accordingly.

Drag the asset into the hierarchy. Assign the target (gameobject), assign a default camera and activate.

Choose the CinematicPerspectives GameObject in the Hierarchy. The inspector will change to the corresponding component.

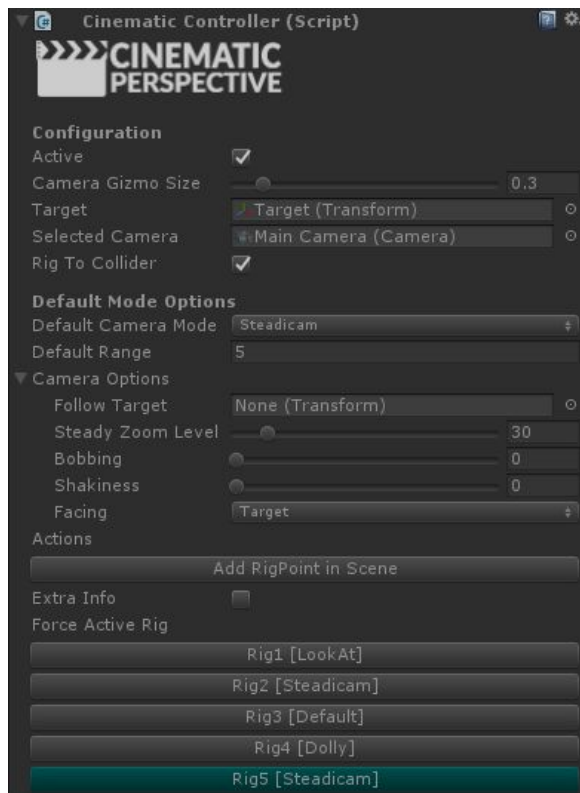


The whole component can be activated or deactivated from the “Active” toggle. The component will not activate if a Target and a

Camera are not previously assigned. The “Camera Gizmo Size” is for better viewing the gizmo in the rig. Change it if desired.

Then a default target and a default camera need to be assigned. In all rigs this camera can be overridden and in steadicam rigs the target can too.

Also, in playmode, the inspector will show another section at the end, Force Active Rig:



This section will allow to choose and “force” a particular rig directly from the inspector and **also to monitor which rig is rendering at the present time.**

Adding Rigs

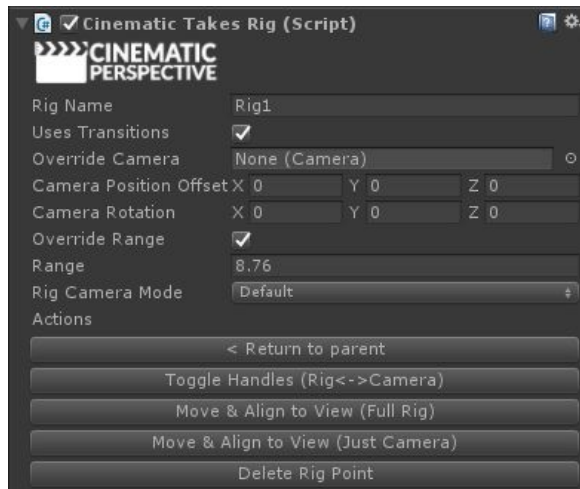
There are two ways. If you have a collider based scenario, you can choose **“Rig To Collider”** option, which will allow you to raycast a green cursor on the desired surface where to place the rig. **By default this icon will be in the Cinematic Perspective position.** To place the cursor please select the “Hand” icon in the top left handle bar. By left clicking on a collider a green dot will appear, which will indicate the placement of the rig once you press the “Add Rig Point in Scene”.

If you don’t want to place the rig on the surface of a collider, deselect the “Rig to Collider Option” and simply place the sceneview camera in the desired alignment and position (see <https://docs.unity3d.com/Manual/SceneViewNavigation.html>). The rig and the camera will be placed where you are set. Finally press the “Add RigPoint in Scene” button and the rig will be added.

To further add new rigs repeat this process.

For usage simplicity we’ve designed the rigs to be gameobjects in the scene. All new rigs added will be placed under the “CameraRigs” empty gameobject for order purposes. The “CameraRigs” gameobject is parented by the “CinematicPerspective” gameobject in the hierarchy.

Once a rig has been added, the selected object in the hierarchy will be the newly added rig.



For better identification the Rig name can be edited.

Since version 0.92b a toggle to activate transitions for this rig is present (see “Transitions” section).

For further convenience, a **particular camera can override the default camera** in any rig. This camera must be set in the “Override Camera” field. This is useful if you needed to have multiple camera scenarios (eg. a camera with and without a depth of field effect).

Once created, the rig position and the camera position are independent. **To switch handles between the rig and the camera** you can press the “Toggle Handles (Rig<->Camera)” button in the corresponding rig or press space bar while having the rig selected in the sceneview or the hierarchy.

The camera position and rotation offset will be shown in the corresponding fields.

The range (spherical trigger radius) shown in the sceneview while the rig is selected, can be overridden for specific rigs by selecting “Override Range” and setting a positive value in the range field.

*When a target enters a particular rig range (the distance between the target and the center of the rig is less than the range value), **the rig will be considered active.***

The selected camera mode (see Introduction section for concept) by default is “Default”. This means that whatever mode has been defined in the “CinematicPerspective” gameobject inspector will be used in this rig.

To Set a Rig

The rigpoint or rig trigger is a gameobject with a spherical gizmo representing the range of the trigger zone. As any gameobject it can be moved in the scene.

As previously explained, the trigger will activate the rig on the target entering the sphere.

Once the target enters the sphere, the selected camera will position itself in the desired offset (or camera position) and activate itself.

If other cameras are present in the scene, they will be ignored, only the selected cameras will be operated by the rig system.

When created, the camera (a gizmo representing the camera position and rotation) will be placed at the center of the rig (of the sphere). To allow better scene customization, **the camera can be placed anywhere**, but it will always remain linked to this rig's spherical trigger.

As previously explained **to switch handles between the rig and the camera** you can press the “Toggle Handles (Rig<->Camera)” button in the corresponding rig or press space bar while having the rig selected in the sceneview or the hierarchy. Both “Move” and “Rotate” handles can be used.

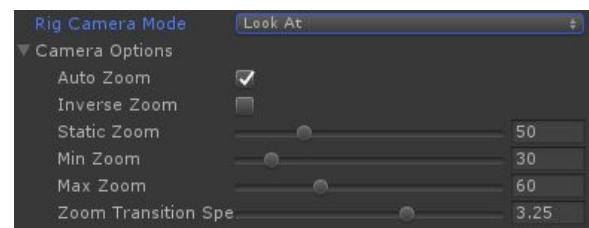
Define a Mode

There are four modes, and each one has its set of particular parameters. If the mode type of the rig is set to default, as explained before, the general mode will apply.

By the way, we have freely reinterpreted the Field of View value in the camera to be Zoom level in most of our modes. We know this is not the case, but we thought that to simplify the concept of FOV, we would imagine it to be a zoom lens without focusing needed or focal distance effects taking place. Please accept our apologies for this.

The “LookAt” Mode

If chosen, this mode will try and follow the target by rotating the camera, but the camera position will remain fixed.



This mode has the following parameters.

The Auto Zoom (should be auto Field of View) is a clamped level variation between the Min Zoom and Max Zoom values which will increase “Zoom” when target increases distance (reaching max zoom level at the range distance). If the inverse toggle is selected, this process will be inverted.

If Auto Zoom is inactive, the static zoom will be used and the FOV will remain constant.

The Zoom transition speed is a speed multiplier to increase the rate at which the FOV will adjust.

Typical use case would be when the target (subject or performer) goes from point a to b and the camera rotates towards the target making focus on him over the scenario.

The “Fixed” Mode

If chosen, this mode will not move or rotate the camera when the rig is selected. Imagine this mode to be similar to the previous mode but zooming is manual and there is no rotation towards target.



This mode has the following parameters.

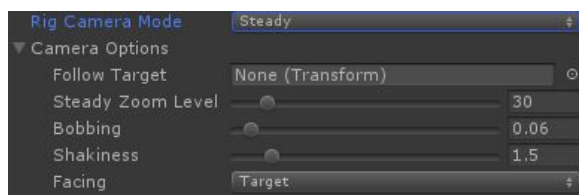
The **Zoom Level** will change the fov in the camera.

Face Target on Enter toggle will rotate the camera towards the target only when the trigger sphere is entered, then the camera will not rotate anymore.

Typical use case would be when the target (subject or performer) goes from point a to b in the scene, but the camera remains constant, showing the whole picture.

The “Steadicam” Mode

If chosen, this mode will move the camera following the target. Also, a different target from the general target can be set for specific effects and creative freedom.



This mode has the following parameters.

The **Follow Target** transform, if defined, will override the CinematicPerspective target.

The **Steady Zoom Level** will change the fov in the camera.

If the **Bobbing** value is more than 0, the camera will be shaken using perlin noise with the **shakiness** being the frequency and the bobbing value being the amplitude. This can be used to simulate the “human shakiness” effect in the steadicam camera.

The **Facing** value will determine the direction camera will face. “Target”, the camera will face the target; “Front”, the camera will look in the target’s “Forward” direction; and “Camera Direction” will persist the camera’s rotation during playmode.

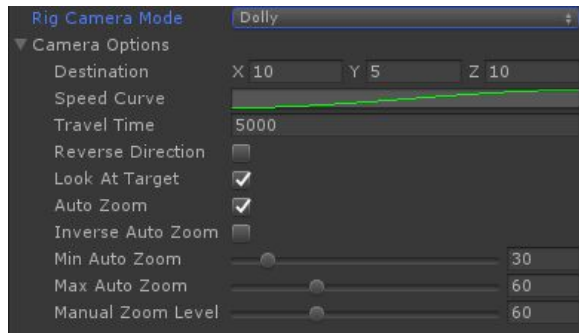
Typical use case would be a follow shot, where the target (subject or performer) goes from point a to b in the scene and is followed by the camera.

The “Dolly” Mode

If chosen, this mode will try and follow a vector defined by the camera position and the vector destination and the camera rotation can follow or not the target.

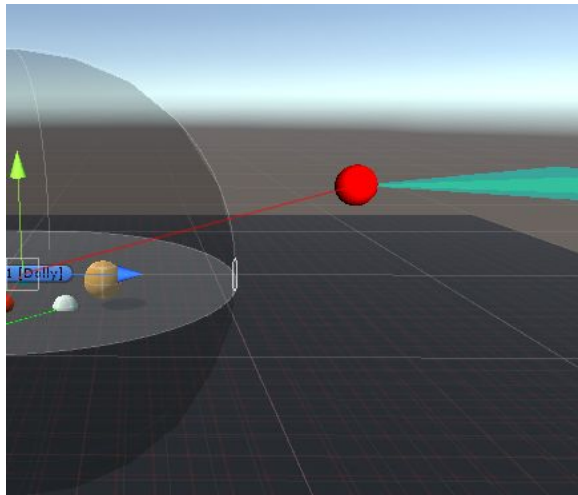
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CINEMATIC PERSPECTIVE



This mode has the following parameters.

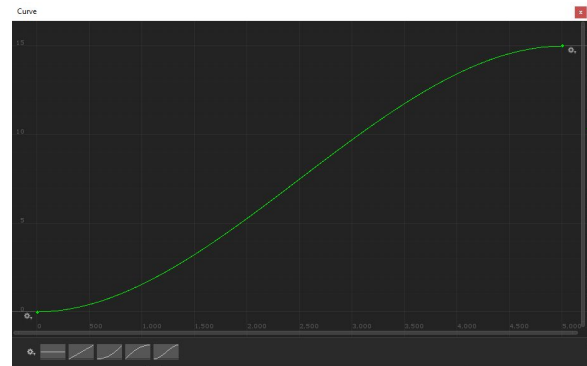
The **Destination** vector is the position at which the camera will arrive when the traversing ends. This vector can be set in the inspector or can be edited in the sceneview with a small red spherical gizmo. The camera's path will be illustrated with a red line. See the following image to identify the gizmo.



The **Travel Time** is the time the camera will take to traverse from the original camera position to the destination vector.

The **Speed Curve** represents camera will move along the path over time, thus allowing a more flexible and realistic approach to the “dolly shot” movement effect.

It is important to note that exceeding the max / min Y values in the curve will make the camera exceed the destination path endings. Also the change in slope will indicate the camera's direction in the path.



If selected, the **Reverse Direction** toggle, will make the destination the origin and the traverse will take place inversely.

The **Look At Target** toggle will make the camera face a particular target while traversing.

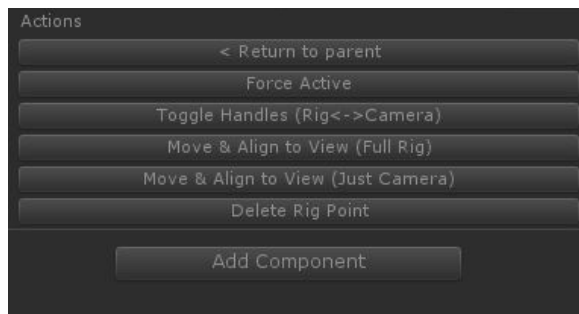
The Auto Zoom and following parameters implement the same functionality as in the “Look At” mode. Please read the previous section.

Typical use case would be to produce images which involve moving the camera toward or away from the target while the camera faces the target (subject or performer), which goes

from point a to b in the scene. This provides a great dynamicity to a shot, which is also known as “dolly shot”.

Additional Actions in Rigs

When a rig is selected, a group of actions (“buttons”) will be shown in the inspector at the bottom of the configuration.



“< Return to Parent” action will simply choose the CinematicPerspective gameobject in the hierarchy and display it’s inspector.

In playmode, a button named “**Force Active**” will force the selected rig to be always the active one over the others, thus forbidding the rig to be replaced upon the target entering other rig ranges and changing rigs.

The “**Toggle Handles (Rig<->Camera)**” button will change the handle selector between the rig and the camera. This is very useful to displace the camera from the rig position. The range can then be in one place and the camera in another.

The “**Move and Align To View**” buttons (for the camera and rig) will move the rig or just the camera to the present scene view position and orientation.

Finally, to **delete a Rig** press the “**Delete Rig Point**” button on the inspected rig o simply delete the rig from the hierarchy.

Transitions

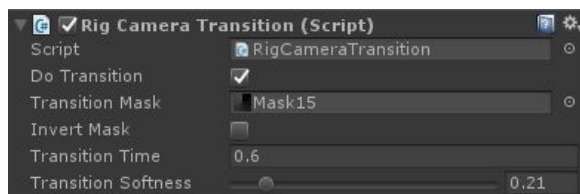
To achieve a more realistic and flexible cinematic film look we’ve also included transitions between rigs in our asset.

Basically, a film transition is a technique used in the post-production process of film editing and video editing by which scenes or shots are combined. Most commonly this is through a normal cut to the next shot (in our case, from one rig to another).

Most films will also include selective use of other transitions, usually to convey a tone or mood, suggest the passage of time, or separate parts of the story.

Since version 0.92b we’ve included two components. One, “CameraTransitions”, that will be automatically added to any camera used by Cinematic Perspective during a transition. By simply selecting this camera in the main “CinematicController” or as an override camera in any rig, this component will be added and removed in runtime. The second is a “RigCameraTransition” component.

Every rig has a toggle included in the main component named “Uses Transitions” (See “Adding Rigs”) that enables the activation of the “RigCameraTransition” component. Upon creation of new rigs this is disabled by default.



All transitions will be applied on leaving the rig. When going from rig A to rig B, the rig A transition will be applied.

The **Do Transition toggle** is basically to disable the selected transition. This can be useful if you want only to activate the transition in certain situations. This is also the correct way to persist the transition settings for the rig in case you want to disable the transition as it will be reset upon disabling and reenabling in the “Uses Transitions” of the “RigCameraTransition” component.

The **Transition Mask** is a 8 bit grayscale single channel (png or jpg) image with a pattern

describing the transition direction. Please take into consideration the mask will be adapted to screen proportions and resolution. The asset is provided with 15 basic transitions, but we encourage you to create your own.

If **no Transition Mask is defined, the transition will simply “dissolve” from one camera to another.**

For reference all masks included in the asset are in the Assets > CinematicPerspective > CinematicTransitions > Masks folder.

The **Invert Mask** toggle will simply invert the transition direction. This allows testing different ways the same transition for flexibility.

The **Transition Time** is a float value that represents the time in seconds the transition will take. This value must be greater than 0 and smaller than the max float val. You must take into consideration that if that if a new trigger is left before the present transition ends, this new transition will be omitted.

The **Transition Softness** is a process that “blurs” the edges in the transition masks application so to achieve better integration between takes. We’ve found that blurring by 0.5 works in most cases for good integrated transitions.

API Reference

The “Cinematic Perspective” asset is provided with a small api that will grow over time. The api is in c# and the basic functionality is as follows:

To manage the component (at this point the object should be attached via script):

```
CinematicController cc = GameObject.Find("GameobjectWithCC").GetComponent<CinematicController >();
```

Get the active status of the component:

```
bool isActive = cc.active;
```

To activate the rendering and functionality of the component

```
if(!cc.active)
{
    cc.SetActive();
}
```

Set the target

```
transform target = cc.setTarget(myTransformTarget);
```

The distance, in game units, between the selected rig and the target

```
CinematicTakesRig rig = cc.getSelectedRig();
float distance = cc.distanceToTarget(rig); // the rig can be omitted and you will get the same distance
```

Force the active rig

```
CinematicTakesRig rig = cc.getSelectedRig(); // Get the present selected rig
SetActiveRig(rig); // Will force the rig to remain active beyond its range
```

Set the transition for the chosen rig

```
rig.usesTransitions = true; // Enables the transition component
rig.CameraTransitions.SetOptions(true, 0.3f, 0.2f, false); // Sets the transition active with various options
(see code)
```

Version History

0.92b Added Transitions (see transition section in the manual). Added “Force Rig” to let the user choose the active rig besides the triggering in the inspector. Improved Gizmos, now all rigs can be seen while one is chosen. Corrected performance issues. Modified User Guide.

0.91b Added setActiveRig() in API and Force Rig Button. Modified User Guide.

0.9b Initial Release

Further Improvements

We are trying to improve our asset and will take into consideration suggestions sent to us. Please don't doubt and contribute any ideas for improvement you may have. FYO some of the future improvements we are considering are add path mode, improving the steadicam mode for object avoidance, include camera destination rotation in dolly mode, adding other type of triggers like cubes, add multiple targets (targeting center of the target array maybe), enable target displacement, add shakiness to target displacement, soft target change on the fly, including fade to colour in transitions and improving the api for full control.